

# Mathematical Methods I

## Differential 1-forms

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### 1. Covectors and 1-forms

A tangent vector describes a direction and a magnitude at a point. A 1-form evaluates that vector and produces a real number in a linear way.

#### Definition 1.1 (Covector and covector field)

Let  $\Omega \subseteq \mathbb{R}^n$  be open. A **covector** at  $p \in \Omega$  is a linear map

$$\omega_p : T_p\Omega \longrightarrow \mathbb{R}.$$

The set of these functionals is the **cotangent space**  $T_p^*\Omega = (T_p\Omega)^*$ , the dual of the tangent space.

A **differential 1-form**  $\omega$  assigns a covector  $\omega_p$  to each point  $p$ , with smooth coefficient functions in the case of a smooth form.

Linearity refers to vectors based at the same point: for  $u_p, v_p \in T_p\Omega$  and  $a, b \in \mathbb{R}$ ,

$$\omega_p(au_p + bv_p) = a\omega_p(u_p) + b\omega_p(v_p).$$

#### Operations with 1-forms

For 1-forms  $\phi, \psi$  and a scalar function  $f$ , define

$$\begin{aligned} (\phi + \psi)_p(v_p) &= \phi_p(v_p) + \psi_p(v_p), \\ (f\phi)_p(v_p) &= f(p)\phi_p(v_p). \end{aligned}$$

The factor  $f$  is evaluated at the base point of the vector; it is not differentiated.

#### Evaluation on a vector field

If  $X$  is a vector field,  $\omega(X)$  is the scalar function

$$(\omega(X))(p) = \omega_p(X(p)).$$

In particular,  $\omega_p(v_p)$  is a number, whereas  $\omega(X)$  is a function of  $p$ .

#### Linearity over functions

For vector fields  $X, Y$  and scalar functions  $f, g$ ,

$$\omega(fX + gY) = f\omega(X) + g\omega(Y).$$

Thus, a 1-form transforms vector fields into functions by pointwise evaluation.

## 2. Coordinate differentials and the dual basis

Let  $x^1, \dots, x^n$  be the Cartesian coordinate functions. We write

$$U_j = \partial_j = \frac{\partial}{\partial x^j}, \quad v_p = \sum_{j=1}^n v^j \partial_j|_p.$$

The superscripts in  $x^i$  and  $v^i$  are indices, not powers.

### Definition 2.1 (Differential of a function)

For a differentiable function  $f : \Omega \rightarrow \mathbb{R}$ , its differential at  $p$  is

$$(df)_p(v_p) = v_p[f].$$

The differentiability of  $f$  guarantees that this expression is linear in  $v_p$ . If  $f$  is smooth,  $df$  is a smooth 1-form. Applied to  $f = x^i$ ,

$$(dx^i)_p(v_p) = v_p[x^i] = v^i.$$

The coordinate 1-form  $dx^i$  selects the  $i$ th component of the vector.

### Proposition 2.2 (Duality of the bases)

At each point,  $\{(dx^1)_p, \dots, (dx^n)_p\}$  is the dual basis of  $\{\partial_1|_p, \dots, \partial_n|_p\}$ . Indeed,

$$(dx^i)_p(\partial_j|_p) = \partial_j|_p[x^i] = \frac{\partial x^i}{\partial x^j}(p) = \delta_j^i, \quad \delta_j^i = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

As an identity between functions, we write  $dx^i(U_j) = \delta_j^i$ .

In Cartesian coordinates, the value  $dx^i(v_p) = v^i$  does not explicitly depend on  $p$  when the vector components are fixed.

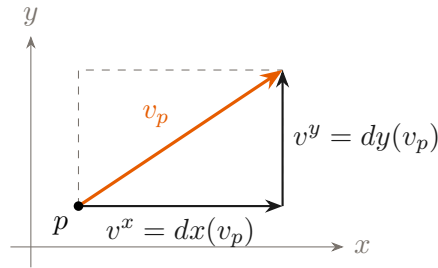


Figure 1: The forms  $dx$  and  $dy$  return the components of the vector based at  $p$ . The horizontal and vertical arrows represent vector components, not the 1-forms themselves.

### 3. Expression and evaluation of a 1-form

#### Proposition 3.1 (Coordinate expression)

Every 1-form can be uniquely expressed as

$$\omega = \sum_{i=1}^n a_i dx^i,$$

where the  $a_i$  are scalar functions. Its coefficients are recovered by evaluating it on the tangent basis:

$$a_i = \omega(U_i).$$

#### Proof and evaluation formula

At a point  $p$ , let  $v_p = \sum_j v^j \partial_j|_p$ . By linearity and duality,

$$\begin{aligned} \omega_p(v_p) &= \left( \sum_i a_i(p) (dx^i)_p \right) \left( \sum_j v^j \partial_j|_p \right) \\ &= \sum_i \sum_j a_i(p) v^j (dx^i)_p(\partial_j|_p) \\ &= \sum_i \sum_j a_i(p) v^j \delta_j^i = \sum_i a_i(p) v^i. \end{aligned}$$

In particular,

$$\omega(U_j) = \sum_i a_i dx^i(U_j) = \sum_i a_i \delta_j^i = a_j.$$

Conversely, defining  $a_i = \omega(U_i)$ , the form  $\sum_i a_i dx^i$  agrees with  $\omega$  on every basis vector and, by linearity, on every vector. This proves the existence and uniqueness of the expression.

#### Evaluation on a vector field

If  $X = \sum_i X^i \partial_i$ , then

$$\omega(X) = \sum_i a_i X^i.$$

Both sides are functions; evaluation at  $p$  gives  $\omega_p(X(p))$ .

#### Example 3.2 (A function and its value at a point)

In  $\mathbb{R}^2$ , let  $\omega = y dx + 2x dy$  and  $X = x \partial_x - \partial_y$ . Then

$$\omega(X) = xy - 2x.$$

For  $p = (1, 3)$ ,  $\omega_p = 3(dx)_p + 2(dy)_p$  and  $X(p) = (1, -1)_p$ , so  $\omega_p(X(p)) = 3 - 2 = 1$ . Moreover,  $\omega(\partial_x) = y$  and  $\omega(\partial_y) = 2x$  recover the coefficients.

## 4. The differential $df$ and exact forms

### Theorem 4.1 (Coordinate expression of the differential)

For a differentiable function  $f : \Omega \rightarrow \mathbb{R}$ ,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i.$$

A 1-form that can be written as  $df$  throughout its domain is called **exact**. The function  $f$  is a potential for that form.

### Proof by evaluation

For any  $v_p = \sum_j v^j \partial_j|_p$ , the definition of the differential gives

$$(df)_p(v_p) = v_p[f] = \sum_i v^i \frac{\partial f}{\partial x^i}(p).$$

On the other hand, using duality,

$$\begin{aligned} \left( \sum_i \frac{\partial f}{\partial x^i}(p) (dx^i)_p \right) (v_p) &= \sum_i \sum_j \frac{\partial f}{\partial x^i}(p) v^j \delta_j^i \\ &= \sum_i \frac{\partial f}{\partial x^i}(p) v^i. \end{aligned}$$

The two forms agree on all tangent vectors and are therefore equal. Equivalently, their coefficients are obtained as

$$df(U_i) = U_i[f] = \frac{\partial f}{\partial x^i}.$$

### Example 4.2 (Computing $df$ )

If  $f(x, y, z) = x^2y + z^2$ , then

$$df = 2xy \, dx + x^2 \, dy + 2z \, dz.$$

For  $p = (1, 2, -1)$  and  $v_p = (2, -1, 3)_p$ ,

$$(df)_p(v_p) = 4(2) + 1(-1) - 2(3) = 1.$$

This gives the directional derivative of  $f$  along the vector  $v_p$ .

### Differential, vector field, and curve

For any vector field  $X$ ,  $df(X) = X[f]$ . If  $\alpha$  is a differentiable curve contained in  $\Omega$ , the chain rule states that

$$(df)_{\alpha(t)}(\alpha'(t)) = \frac{d}{dt}(f(\alpha(t))).$$

Evaluating  $df$  on the velocity measures how  $f$  changes along the curve.

## 5. The operator $d$ : products and compositions

The operator  $d$  transforms a scalar function  $f$  into the 1-form  $df$ . The usual differentiation rules thus become equalities between 1-forms. We assume that the functions are differentiable on their domains.

### Proposition 5.1 (Linearity and the product rule)

For  $f, g : \Omega \rightarrow \mathbb{R}$  and constants  $a, b \in \mathbb{R}$ ,

$$d(af + bg) = a df + b dg, \quad d(c) = 0 \quad (c \text{ constant}).$$

The **Leibniz rule** states that

$$d(fg) = f dg + g df.$$

The factors  $f$  and  $g$  multiply 1-forms. At a point  $p$ , this equality means

$$(d(fg))_p = f(p)(dg)_p + g(p)(df)_p.$$

### Why the product rule holds

For any tangent vector  $v_p$ , we apply the definition of the differential and the product rule for directional derivatives:

$$\begin{aligned} (d(fg))_p(v_p) &= v_p[fg] \\ &= f(p) v_p[g] + g(p) v_p[f] \\ &= f(p)(dg)_p(v_p) + g(p)(df)_p(v_p). \end{aligned}$$

Since both 1-forms give the same result on every  $v_p$ , they are equal. Linearity is justified in the same way.

### Example 5.2 (Product of two functions)

Let  $f(x, y) = x^2$  and  $g(x, y) = y$ . First, we compute

$$df = 2x dx, \quad dg = dy.$$

Then we apply the rule:

$$\begin{aligned} d(x^2 y) &= x^2 d(y) + y d(x^2) \\ &= x^2 dy + 2xy dx. \end{aligned}$$

This gives the term corresponding to  $x^2 y$  in Example 4.2. Since  $d(z^2) = 2z dz$ , linearity gives

$$d(x^2 y + z^2) = 2xy dx + x^2 dy + 2z dz.$$

In  $f dg$ , the function multiplies the form:  $(f dg)_p(v_p) = f(p)(dg)_p(v_p)$ . Two differentials are not multiplied together;  $d(fg)$  is a 1-form.

## 5. The operator $d$ (continued)

### Proposition 5.3 (Chain rule)

Let  $f : \Omega \rightarrow \mathbb{R}$  and  $h : A \rightarrow \mathbb{R}$ , where  $A \subseteq \mathbb{R}$  is open and  $f(\Omega) \subseteq A$ . Then

$$d(h \circ f) = (h' \circ f) df = h'(f) df.$$

The outer function  $h$  is differentiated, its derivative is evaluated at  $f$ , and the result is multiplied by the differential of the inner function.

### Proof by evaluation

For every tangent vector  $v_p$ , the ordinary chain rule gives

$$(d(h \circ f))_p(v_p) = v_p[h \circ f] = h'(f(p)) v_p[f] = h'(f(p)) (df)_p(v_p).$$

This proves the equality of the 1-forms.

### Example 5.4 (A composition step by step)

For  $F(x, y) = \sin(x^2y)$ , the inner function is  $f(x, y) = x^2y$  and the outer function is  $h(u) = \sin u$ . We have

$$h'(u) = \cos u, \quad df = 2xy dx + x^2 dy.$$

Therefore,

$$dF = \cos(x^2y)(2xy dx + x^2 dy).$$

The factor  $\cos(x^2y)$  multiplies the entire 1-form  $df$ .

### Composition with several inner functions

If  $F = H(f_1, \dots, f_m)$ , where  $H$  is differentiable on an open set containing the image of  $(f_1, \dots, f_m)$ , then

$$dF = \sum_{a=1}^m \frac{\partial H}{\partial u^a}(f_1, \dots, f_m) df_a.$$

For example, for  $F = e^{xy} + z^2$ , take  $H(u, v) = e^u + v^2$ ,  $f_1 = xy$ , and  $f_2 = z$ . Thus,

$$dF = e^{xy} d(xy) + 2z dz = e^{xy}(y dx + x dy) + 2z dz.$$

### Useful consequences

$$d(f^k) = k f^{k-1} df \quad (k = 1, 2, \dots), \quad d(e^f) = e^f df.$$

Combining the product and composition rules, where  $g \neq 0$ ,

$$d\left(\frac{f}{g}\right) = \frac{1}{g} df - \frac{f}{g^2} dg = \frac{g df - f dg}{g^2}.$$

## 6. Indices and coordinate changes

The components of a vector and a 1-form change with the coordinates, although the geometric object and its evaluation remain the same.

### Einstein summation convention

An index repeated once as a superscript and once as a subscript indicates summation:

$$X = X^i \partial_i, \quad \omega = \omega_i dx^i, \quad \omega(X) = \omega_i X^i.$$

The repeated index is a **dummy index**: it may be renamed without changing the expression. Free indices must match on both sides of an equality.

The vector components  $X^i$  are **contravariant**; the components  $\omega_i$  of a 1-form are **covariant**. The tangent basis  $\partial_i$  and the cotangent basis  $dx^i$  transform so as to compensate for the change in their respective components.

### General rules without matrix ambiguity

Let  $x = (x^1, \dots, x^n)$  and  $u = (u^1, \dots, u^n)$  be two local coordinate systems, with invertible change of coordinates  $x = x(u)$ . Set

$$J^i_a = \frac{\partial x^i}{\partial u^a}, \quad (J^{-1})^a_i = \frac{\partial u^a}{\partial x^i}.$$

### Transformation of bases and components

By the chain rule,

$$\begin{aligned} \partial_{u^a} &= \frac{\partial x^i}{\partial u^a} \partial_{x^i}, & du^a &= \frac{\partial u^a}{\partial x^i} dx^i, \\ X_u^a &= \frac{\partial u^a}{\partial x^i} X_x^i, & \omega_{u,a} &= \frac{\partial x^i}{\partial u^a} \omega_{x,i}. \end{aligned}$$

Writing each list as a column, these identities become

$$\partial_u = J^T \partial_x, \quad du = J^{-1} dx, \quad X_u = J^{-1} X_x, \quad \omega_u = J^T \omega_x.$$

### Invariance of evaluation

Contracting a covariant index with a contravariant one gives

$$\omega_{u,a} X_u^a = \frac{\partial x^i}{\partial u^a} \omega_{x,i} \frac{\partial u^a}{\partial x^j} X_x^j = \delta_j^i \omega_{x,i} X_x^j = \omega_{x,i} X_x^i.$$

An index repeated twice as a superscript or twice as a subscript does not represent a contraction under this convention. For example, the inner product is written  $g_{ij} A^i B^j$ ; in an orthonormal Cartesian basis it can be simplified to  $\sum_i A^i B^i$ , because there  $g_{ij} = \delta_{ij}$ .

## 7. Example: polar coordinates

Consider  $x = r \cos \theta$ ,  $y = r \sin \theta$ , with  $r > 0$  and a smooth local choice of the angle. The origin lies outside this chart.

### Jacobian and inverse matrix

With  $J = \partial(x, y)/\partial(r, \theta)$ ,

$$J = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad J^{-1} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta/r & \cos \theta/r \end{pmatrix}.$$

We have  $\det J = r \neq 0$ .

### Tangent basis and vector components

$$\begin{aligned} \partial_r &= \cos \theta \partial_x + \sin \theta \partial_y, \\ \partial_\theta &= -r \sin \theta \partial_x + r \cos \theta \partial_y, \\ V^r &= \cos \theta V^x + \sin \theta V^y, \\ V^\theta &= -\frac{\sin \theta}{r} V^x + \frac{\cos \theta}{r} V^y. \end{aligned}$$

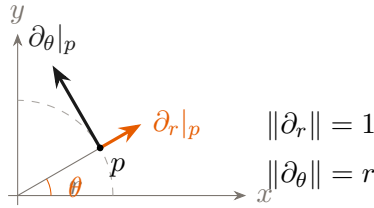


Figure 2: Polar coordinate basis, illustrated at  $r = 2$ . The vector  $\partial_\theta$  is tangent to the circle and has length  $r$ ; it is not generally a unit vector.

### Cotangent basis and 1-form components

$$\begin{aligned} dr &= \cos \theta dx + \sin \theta dy, \\ d\theta &= -\frac{\sin \theta}{r} dx + \frac{\cos \theta}{r} dy, \\ \omega_r &= \cos \theta \omega_x + \sin \theta \omega_y, \\ \omega_\theta &= -r \sin \theta \omega_x + r \cos \theta \omega_y. \end{aligned}$$

These formulas verify that  $dr(\partial_r) = d\theta(\partial_\theta) = 1$  and  $dr(\partial_\theta) = d\theta(\partial_r) = 0$ . With the columns and Jacobian defined above, the correct relation is

$$\begin{pmatrix} dr \\ d\theta \end{pmatrix} = J^{-1} \begin{pmatrix} dx \\ dy \end{pmatrix}.$$

## 8. Verification in both coordinate systems

### Example 8.1 (A 1-form and a vector)

Consider

$$\omega = x dx + y dy, \quad V = \partial_x.$$

In Cartesian coordinates,  $\omega(V) = x$ .

In polar coordinates, the components of the form are

$$\begin{aligned}\omega_r &= x \cos \theta + y \sin \theta = r, \\ \omega_\theta &= -rx \sin \theta + ry \cos \theta = 0.\end{aligned}$$

Therefore,  $\omega = r dr$ . The components of the same vector are

$$V^r = \cos \theta, \quad V^\theta = -\frac{\sin \theta}{r}.$$

Evaluation in the polar basis gives

$$\omega(V) = \omega_r V^r + \omega_\theta V^\theta = r \cos \theta = x.$$

The results agree: evaluation does not depend on the coordinates.

### The same form as an exact differential

Define

$$f(x, y) = \frac{x^2 + y^2}{2}.$$

In Cartesian coordinates,  $df = x dx + y dy$ . In polar coordinates, the same function is expressed as  $f = r^2/2$ , so

$$df = r dr.$$

The two expressions describe the same 1-form. The potential  $f$  exists on all of  $\mathbb{R}^2$ , although polar coordinates are used only locally away from the origin.

### Differential and gradient

The differential  $df$  is a covector. In Euclidean space, the inner product relates it to the gradient vector:

$$(df)_p(v_p) = \nabla f(p) \cdot v.$$

This identification uses the metric; the definition of  $df$  as a linear functional does not require it.